

Debugging the Continuous-Rearrangement Proof

What survives, what fails, and the quantitative route that remains

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Abstract

This note audits `faber_krahn_rearrangement_analysis.tex`. The conclusion is not that continuous rearrangement is useless; the conclusion is that the current Brock/Steiner schedule proof asks the wrong object to do the sharp endpoint job. The finite-time dense-schedule claim is false, the stated one-dimensional quantitative Polya–Szegő theorem is false as written, the multidimensional Fubini transition ignores transverse derivatives, and the centre-drift lemma is false. The repair is to replace the one-shot directional story by the Plan 2 continuous *level-set* rearrangement: Talenti’s level-set identity gives the exact budget

$$\delta_T(\Omega) \simeq \int (D_H + D_I),$$

where D_I controls distance of many levels from balls and D_H controls the kinetic cost of moving between levels. A weighted metric trace then gives a good near-boundary level; the boundary-layer transfer returns the estimate to Ω . This is the route in the directory with the correct $H^{1/2}$ /quadratic scale. It is fully quantitative once the explicitly isolated remaining lemmas are proved with constants.

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1 The target and the non-negotiable scale

Normalize $|\Omega| = \omega_n$, and write $u = u_\Omega$. The sharp Saint–Venant/Faber–Krahn target is

$$\mathcal{A}(\Omega)^2 \leq C(n) \delta_T(\Omega). \tag{1}$$

The rearrangement bookkeeping begins with the exact splitting

$$\delta_T(\Omega) = \underbrace{\left(\int |\nabla u|^2 - \int |\nabla u^*|^2 \right)}_{\mathcal{D}_{\text{rr}}(\Omega)} + \underbrace{(T(B) - J(u^*))}_{\mathcal{R}_{\text{rad}}(\Omega)}. \quad (2)$$

This identity is sound. The radial profile term is harmless: by strict concavity of the torsion functional on the ball,

$$\mathcal{R}_{\text{rad}}(\Omega) = \int_B |\nabla(u^* - u_B)|^2.$$

The question is how to convert $\mathcal{D}_{\text{rr}}$, or a refined form of it, into geometric asymmetry at the sharp quadratic scale.

2 Where the current proof breaks

2.1 Finite-time dense schedules do not see all directions

The theorem called “Lower Bound for Dense Schedules” in the current rearrangement-analysis note is false for finite time. If the step size is fixed and $T < \Delta_0$, the flow uses only one Steiner direction. Any non-radial function already symmetric in that direction has zero dissipation on $[0, T]$ but positive distance from the radial cone. More generally, a finite interval samples only finitely many directions; it never samples a dense subset of $\mathbb{S}^{n-1}$.

Even if the statement is weakened to an infinite-time convergence statement, the proof is not quantitative. The compactness argument would produce at best a modulus depending on a compact class, the schedule, and the topology. It does not produce the computable quadratic constant needed for (1). The missing theorem would have to be a genuine angular spectral-gap estimate for averaged instantaneous dissipation, not compactness.

2.2 The stated one-dimensional theorem is false

The note states a scale-free estimate

$$\int |g'|^2 - \int |(g^*)'|^2 \geq c_0 \inf_z \|g - g^*(\cdot - z)\|_{L^2(\mathbb{R})}^2 \quad (3)$$

for all compactly supported H^1 functions on the line. This cannot hold with a universal c_0 . If $g_L(x) = g(x/L)$, then the left side scales like L^{-1} , while the right side scales like L . The ratio is L^{-2} .

There is a second, structural error: non-symmetric does not imply that the level sets $\{g > t\}$ have multiple intervals on a positive set of heights. A single asymmetric tent function has one interval at every height, but is not symmetric decreasing up to translation. The multi-interval Cauchy–Schwarz estimate is correct but incomplete.

2.3 The missing one-dimensional term is centre drift

For one regular interval $\{g > t\} = (a(t), b(t))$, set

$$p(t) = \frac{1}{g'(a(t))} > 0, \quad q(t) = \frac{-1}{g'(b(t))} > 0.$$

Then $a' = p$, $b' = -q$, the length $\ell = b - a$ satisfies $-\ell' = p + q$, and the centre $c = (a + b)/2$ satisfies

$$c' = \frac{p - q}{2}.$$

The exact one-interval Polya–Szegő level loss is

$$\frac{1}{p} + \frac{1}{q} - \frac{4}{p+q} = \frac{(p-q)^2}{pq(p+q)} = \frac{4|c'|^2}{pq(p+q)}. \quad (4)$$

Thus a one-interval asymmetric slice is not an obstruction by itself; it is controlled through variation of the level-centre $c(t)$, provided the weights are non-degenerate on the height interval under study. This is precisely why a correct proof needs profile-gap and bad-density control. The current proof underclaims here by treating multiplicity as the only source of dissipation, and overclaims by pretending the multiplicity term controls all asymmetry.

2.4 The multidimensional Fubini step is not exact

For a fixed direction ν , slicing gives an identity for the ∂_ν -part of the gradient, not for the full Dirichlet energy:

$$\int_{\mathbb{R}^n} |\partial_\nu u|^2 - \int_{\mathbb{R}^n} |\partial_\nu u^{*\nu}|^2 = \int_{\nu^\perp} \left(\int |(u_{y,\nu})'|^2 - \int |((u_{y,\nu})^*)'|^2 \right) dy.$$

The full difference

$$\int |\nabla u|^2 - \int |\nabla u^{*\nu}|^2$$

also involves transverse derivatives. Brock’s theorem controls the full Dirichlet energy under continuous Steiner symmetrization, but it is not a plain Fubini identity. Any proof that averages one-dimensional slice losses must either use Brock’s full formula or switch to a level-set identity where all terms are accounted for.

2.5 The centre-drift quarantine lemma is false

The asserted estimate

$$\int_{\nu^\perp} |\bar{s}_\nu(y) - x_0 \cdot \nu|^2 M_\nu(y) dy \lesssim \int |\nabla u|^2 - \int |\nabla u^{*\nu}|^2$$

is false. Take a function whose slices in the s -direction are translated copies of a fixed symmetric profile, with the translation depending on y , and with negligible y -transition cost in separated components. The one-direction Steiner gap can be zero on each slice, while the variance of slice centres is positive. The proof also uses an infinite factor $\int_{\mathbb{R}} s^2 ds$, an unavailable Hopf lower bound for all slice masses, and a Poincaré estimate for $u - u^{*\nu}$ that is not true in the stated form.

3 The repaired quantitative object

The route in the directory that has the right scale is not the finite directional Brock schedule. It is the Talenti/Plan 2 continuous level-set rearrangement. Let

$$E_t := \{u > t\}, \quad m(t) := |E_t|.$$

For a.e. regular level define

$$D_H(t) := \left(\int_{\partial^* E_t} \frac{1}{|\nabla u|} d\mathcal{H}^{n-1} \right) \left(\int_{\partial^* E_t} |\nabla u| d\mathcal{H}^{n-1} \right) - P(E_t)^2,$$

$$D_I(t) := P(E_t)^2 - n^2 \omega_n^{2/n} m(t)^{2-2/n}.$$

The first term is the Holder/Cauchy–Schwarz loss; the second is the isoperimetric loss of the level set. The proved level-set identity in the Plan 2 material is

$$\frac{2\delta_T(\Omega)}{\omega_n} = \frac{1}{\omega_n} \int_0^{\|u\|_\infty} \frac{D_H(t) + D_I(t)}{n^2 \omega_n^{2/n} m(t)^{1-2/n}} dt. \quad (5)$$

This is the quantitative replacement for the informal statement “the rearrangement flow dissipates energy”. It is exact, levelwise, and separates the two mechanisms that the broken proof conflated:

- D_I says that many levels are close to balls;
- D_H says that the level flow has controlled normal-speed oscillation, hence controlled kinetic motion between nearby levels.

4 The debugged route to the sharp theorem

Parametrize levels by volume radius:

$$|E_\rho| = \omega_n \rho^n, \quad F_\rho := \rho^{-1} E_\rho.$$

Work in the quotient metric space

$$d_{\mathcal{X}}([A], [C]) = \inf_{z \in \mathbb{R}^n} |A \Delta (C + z)|.$$

The correct sharp route is the following.

Step 1. Exact budget. Use (5) to obtain, on any fixed bulk window $\rho \in [\rho_*, 1]$,

$$\int_{\rho_*}^{\rho_\delta} (D_H(t(\rho)) + D_I(t(\rho))) d\mu(\rho) \leq C(n, \rho_*) \delta_T(\Omega),$$

where $d\mu(\rho) = (-t'(\rho)) d\rho$.

Step 2. Distance on many levels from D_I . On levels with small D_I , a same-centre strong quantitative isoperimetric estimate gives

$$d_{\mathcal{X}}(F_\rho, B_1)^2 \leq C D_I(t(\rho)).$$

This is where the strong Fusco–Julin type input belongs.

Step 3. Kinetic control from D_H . For a Borel centre field z_ρ , the desired metric first variation is

$$|\dot{F}_\rho|_{\mathcal{X}} \leq \rho^{-n} \inf_{a \in \mathbb{R}^n} \int_{\partial^* E_\rho} |V_\rho - H_{z_\rho, \rho} - a \cdot \nu_\rho| d\mathcal{H}^{n-1},$$

where

$$V_\rho = \frac{-t'(\rho)}{|\nabla u|}, \quad H_{z_\rho, \rho}(x) = \frac{(x - z_\rho) \cdot \nu_\rho(x)}{\rho}.$$

The self-truncated velocity lemma in the directory shows that the low-gradient part is paid for by D_H ; no Hopf lower bound is needed. Together with the same-centre homothetic trace this gives

$$|\dot{F}_\rho|_{\mathcal{X}}^2 \leq C(D_H(t(\rho)) + D_I(t(\rho)))$$

on good levels.

Step 4. Weighted trace. The profile-gap lemma gives a density lower bound for $d\mu$ outside a bad set of Lebesgue measure $O(\delta_T)$ near

$$\rho_\delta = 1 - \kappa \sqrt{\delta_T(\Omega)}.$$

Combining Markov on the distance term with Cauchy–Schwarz on the kinetic term over an order-one window gives

$$d_{\mathcal{X}}(F_{\rho_\delta}, B_1)^2 \leq C \delta_T(\Omega),$$

or the good-endpoint variant gives the same estimate at $\hat{\rho} \in [\rho_\delta - C\delta_T, \rho_\delta]$.

Step 5. Boundary-layer transfer. Since $E_{\hat{\rho}} \subset \Omega$ and

$$|\Omega \setminus E_{\hat{\rho}}| = \omega_n(1 - \hat{\rho}^n) \leq C \sqrt{\delta_T(\Omega)},$$

the elementary symmetric-difference transfer gives

$$\mathcal{A}(\Omega) \leq \mathcal{A}(E_{\hat{\rho}}) + C \sqrt{\delta_T(\Omega)}.$$

The metric trace gives

$$\mathcal{A}(E_{\hat{\rho}})^2 \leq C \delta_T(\Omega).$$

Squaring the transfer inequality proves the sharp estimate

$$\mathcal{A}(\Omega)^2 \leq C \delta_T(\Omega).$$

This route is a continuous rearrangement proof in the right sense: the “time” is the level/volume radius, the exact energy budget is (5), and the proof is an action-plus-trace argument. It is not the finite-time dense-direction Brock proof.

5 What remains to make it fully quantitative

The remaining tasks are now clean. If each item below is proved with constants, then all constants in the final theorem are computable by composition.

Problem 5.1 (Same-centre strong isoperimetric input). For every finite-perimeter set E , prove or cite in exactly this form that there is one centre z_E such that

$$\left(\frac{|E \Delta B_E(z_E)|}{|E|} \right)^2 + \frac{1}{P(B_E)} \int_{\partial^* E} \left| \nu_E - \frac{x - z_E}{|x - z_E|} \right|^2 d\mathcal{H}^{n-1} \leq C(n) \delta_{\text{iso}}(E).$$

The strong theorem of Fusco–Julin controls Fraenkel asymmetry and boundary oscillation, but the route needs the *same* centre in both terms. If the cited statement is only separately minimized, a centre comparison lemma must be added.

Problem 5.2 (Finite-perimeter metric upper gradient). Upgrade the smooth first-variation formula for $\rho \mapsto [\rho^{-1} E_\rho]$ to finite-perimeter level sets:

$$|\dot{F}_\rho|_{\mathcal{X}} \leq \rho^{-n} \inf_{a \in \mathbb{R}^n} \int_{\partial^* E_\rho} |V_\rho - H_{z_\rho, \rho} - a \cdot \nu_\rho| d\mathcal{H}^{n-1}$$

for a.e. ρ . Lower semicontinuity is the wrong direction; the missing statement is a genuine limsup recovery theorem for the boundary action, or a direct finite-perimeter deformation estimate.

Problem 5.3 (Bad-level kinetic control). Control the kinetic action on the bad isoperimetric or bad-density levels, or avoid transporting through those levels. The good-endpoint variant shows that the final level may be chosen in $G_I \cap G_\tau$, but it does not remove the need for kinetic control when propagating the Markov level to that endpoint. A measure estimate for the bad-density set is not enough; one needs the right $L^2(d\rho)$ action scale.

Problem 5.4 (Globalization). After the bounded result is proved, globalize it by a quantitative bounded reduction. The existing final assembly uses the constructive BDV truncation lemmas and a suboptimal rearrangement bound, not BDV’s final sharp stability theorem. If the aim is to avoid every BDV input, those truncation lemmas must be reproved independently with explicit constants.

6 Verdict

The proof in `faber_krahn_rearrangement_analysis.tex` cannot be patched by minor edits. Its main theorem-level claims are false as stated. The useful idea inside it is the dissipation picture, but the fully quantitative implementation must be the Plan 2 level-set action route, not the finite dense-direction schedule.

The sharp theorem is therefore reduced to Problems 5.1–5.3 in the bounded setting, followed by Problem 5.4 if one wants the global theorem. This is a genuine quantitative route: no compactness modulus is needed in the final trace argument, and all constants are computable once the named inputs are proved quantitatively.

References

- [1] N. Fusco and V. Julin, *A strong form of the quantitative isoperimetric inequality*, Calc. Var. Partial Differential Equations 50 (2014), 925–937; arXiv:1111.4866.
- [2] `Manuscript/plan2_levelset_identity.tex`. Exact level-set identity, profile-gap estimate, and superlevel transfer.
- [3] `Plan 2/WIP/WIP_WeightedMetricTrace.tex`. Conditional integrated action and weighted metric trace.
- [4] `Plan 2/WIP/WIP_MetricFramework.tex`. Metric first variation and the finite-perimeter upper-gradient gap.
- [5] `Plan 2/WIP/WIP_TrimmedVelocityRepair.tex`. Self-truncated velocity defect controlled by D_H .